
Window t
FiLM-CNN encoder
$\mathbf{f}_t \in \mathbb{R}^F$
ApEn per channel
Linear$(F, 32)$
LeakyReLU
Dropout(0.2)
Linear$(32, 126)$
zero-init
split into $(\Delta\boldsymbol{\gamma}_i, \boldsymbol{\beta}_i)_{i=1..5}$
$\boldsymbol{\gamma}_i = 1 + \Delta\boldsymbol{\gamma}_i$, 63 channels
FiLM generator
$\boldsymbol{\gamma}_1, \boldsymbol{\beta}_1$
$\boldsymbol{\gamma}_2, \boldsymbol{\beta}_2$
$\boldsymbol{\gamma}_3, \boldsymbol{\beta}_3$
$\boldsymbol{\gamma}_4, \boldsymbol{\beta}_4$
$\boldsymbol{\gamma}_5, \boldsymbol{\beta}_5$
$\mathbf{x}_t \in \mathbb{R}^{23\times256}$
EEG window (1 s, 23 ch)
23×256
FiLM block 1
$23 \to 18$ ch, k=6
18×125
FiLM block 2
$18 \to 15$ ch, k=5
15×60
FiLM block 3
$15 \to 10$ ch, k=4
10×28
FiLM block 4
$10 \to 10$ ch, k=4
10×12
FiLM block 5
$10 \to 10$ ch, k=4
10×4
Flatten
10×4
inside each FiLM block i
$\mathbf{a}_{i-1}$
Conv1d
k_i, no bias
BatchNorm1d
affine = False
u
FiLM
$\boldsymbol{\gamma}_i \odot \mathbf{u} + \boldsymbol{\beta}_i$
LeakyReLU
slope 0.01
Dropout1d
$p = 0.1$
MaxPool1d
size 2
$\mathbf{a}_i$
$\mathbf{z}_t \in \mathbb{R}^{40}$
Recurrent classifier
delay: keep $\mathbf{h}_t$ for the next window
h_{t-1}
RNN cell (16 units)
$\mathbf{h}_t = \tanh(W_z\mathbf{z}_t + W_h\mathbf{h}_{t-1} + \mathbf{b})$
h_t
Dropout
$p = 0.3$
Linear
$16 \to 2$
$\hat{\mathbf{y}}_t \in \mathbb{R}^2$